

Homework 3 Questions 1-23 Due 9/30/26

1.) 10111101 in Hex is:

$$\begin{array}{r} 1011 \quad 1101 \\ 2^3 \quad 2^2 \quad 2^1 \quad 2^0 \end{array} \quad \boxed{BD}$$

$$2^3 + 2^2 + 2^1 + 2^0 = 8 + 4 + 2 + 1 = 11$$

$$11 = B \quad 8 + 4 + 1 = 13$$

$$D = 13$$

2.) 1011110100000001 as an unsigned decimal

$$2^0 + 2^8 + 2^{10} + 2^{11} + 2^{12} + 2^{13} + 2^{15} = \boxed{48,385}$$

3.) 1011110100000001 as a signed decimal

$$48,385 - 65,536 = \boxed{-17,151}$$

4.) 8000 in 16 Bit Hex:

$$16 \overline{) 8000}$$

$$\boxed{0X1F40}$$

$$16 \overline{) 500} \rightarrow 1F40$$

$$16 \overline{) 31} \quad 4$$

$$16 \overline{) 1} \quad 15 = F$$

5.) 8000 in 16 Bit (signed) Hex

$$\hookrightarrow \boxed{0X1540}$$

6.) -11 (signed) Hex = 65,536 - 11 = 65,525

$$16 \overline{) 65,525}$$

$$16 \overline{) 4095} \quad 5$$

$$16 \overline{) 255} \quad 15$$

$$16 \overline{) 15} \quad 15$$

$$\overline{) 0} \quad 15$$

$$= \boxed{0XFFFS}$$

7.) -32717 in Hex?

$$65,536 - 32,717 = 32,819$$

$$\begin{array}{r} 16 \overline{) 32,819} \\ 16 \overline{) 2091} \quad 3 \\ 16 \overline{) 128} \quad 3 \\ 16 \overline{) 8} \quad 0 \\ \underline{0} \quad 8 \end{array} \quad \boxed{0x8033}$$

8.) Hex FAC3 Binary? =

$$0xFAC3 = 1111 \ 1010 \ 1100 \ 0011$$

$$F=15$$

$$A=10$$

$$C=12$$

9.) FAC3 unsigned Decimal

$$0xFAC3 \rightarrow 3(16^0) + 12(16^1) + 10(16^2) + 15(16^3)$$

$$\begin{array}{l} C=12 \\ A=10 \end{array}$$

$$\rightarrow \boxed{64,195}$$

10.) 0xFAC3 as a signed decimal

$$64,195 - 65,536 = \boxed{-1,341}$$

11.) 0x0064 in Binary?

$$0000 \ 0000 \ 0110 \ 0010$$

12.) 0x0064 unsigned Decimal?

$$\begin{array}{r} 16 \\ \times 6 \\ \hline 46 \end{array} \quad \begin{array}{l} 4(16^0) + 6(16^1) + 0(16^2) + 0(16^3) \\ 4(1) + 6(16) = 4 + 96 = \boxed{100} \end{array}$$

13.) 0x0064 signed decimal?

$$\boxed{100}$$

→ leadin 0 which means
(+) value

14.) $0x8000$ in Binary%

0100 0000 0000 0000

15.) $0x8000$ unsigned Decimal%

$$0(16^0) + 0(16^1) + 0(16^2) + 8(16^3)$$

$$= \underline{\underline{32,768}}$$

$$16^2 = 256$$

$$\begin{array}{r} 256 \\ \times 16 \\ \hline 4096 \end{array}$$

$$\begin{array}{r} 74 \\ 4096 \\ \times 8 \\ \hline 32768 \end{array}$$

16.) $0x8000$ signed%

$$65,536 - 32,768 = \underline{\underline{-32,768}}$$

17.) - If you had 20-Bit registers
smallest signed decimal value%

$$-(2^{19}) = \underline{\underline{-524,288}}$$

18.) largest%

$$2^{19} - 1 = \underline{\underline{524,287}}$$

19.) Modular
sum%

16 Bit Hex $\rightarrow$

$$\begin{array}{r} 3511 \\ + 4FFC \\ \hline 850D \end{array}$$

$$\begin{array}{l} C=12 \\ F=15 \end{array}$$

$$\begin{array}{r} 15 \quad 21 \\ +6 \quad -16 \\ \hline 21 \quad 5 \end{array}$$

$\rightarrow$ 850D

20.) Saturated
sum $\rightarrow$

$$\begin{array}{r} 3511 \\ + 4FFC \\ \hline 850D \end{array}$$

$$\begin{array}{l} C=12 \\ F=15 \end{array}$$

$\rightarrow$ Saturated: FFFF

21.)

$$\begin{array}{r} 0x \quad 3511 \\ 0x \quad 4FFC \\ \hline 850D \end{array}$$

$$C=12$$

$$\begin{array}{r} 3511 \rightarrow 0011 \quad 0101 \quad 0001 \quad 0001 \\ 4FFC \rightarrow + 0100 \quad 1111 \quad 1111 \quad 1100 \\ \hline 10101 \quad 0000 \quad 1101 \end{array}$$

Carry% NO!

22.)

overflow% yes!

2 positives results
in a negative